LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel contamination with water, foreign particles, other types of fuel, micro biological growth, contamination detection and control
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel contamination with water, foreign particles, other types of fuel, micro biological growth, contamination detection and control
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of fuel water separator operation and visual inspection of fuel strainer elements for particulate contamination; Instructor supervision required for safety and equipment handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary sources of fuel contamination in aircraft systems?
(b) Why is water separation critical before fuel reaches the engine?
(c) How does micro-biological growth affect fuel system integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of fuel contamination sources • Water contamination mechanisms and effects • Foreign particle ingress and filtration • Micro biological growth conditions • Fuel system component requirements	L & Ds	PPT, IB	15 min
(b)	• Detection methods for water and particles • Sampling procedures and visual inspection • Filter maintenance and replacement intervals • Control measures for microbial growth • Impact of wrong fuel types on engines	L & Ds	PPT, PDF	20 min
(c)	• Standard operating procedures for fuel checks • Reporting contamination incidents • Preventive maintenance strategies • Emergency handling of contaminated fuel • Record keeping for fuel quality logs	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between water accumulation and microbial growth in fuel tanks.
(b) Describe the steps taken to isolate and drain contaminated fuel from the system.

12. Home Assignment and Follow Up	Prepare a brief note on how the fuel system design in LP 02 (Hydraulic system) principles of cleanliness apply to preventing cross-contamination between hydraulic and fuel lines.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min